

Khulna University of Engineering & Technology (KUET)

Department of Computer Science and Engineering

Machine Learning Project Report

A Machine Learning Framework for Predicting and Ranking the Quality of University Websites

Course Information

Course Title	Machine Learning Laboratory
Course No.	CSE 4112
Department	Computer Science and Engineering
University	Khulna University of Engineering & Technology (KUET)

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Submission Date: September 2, 2026

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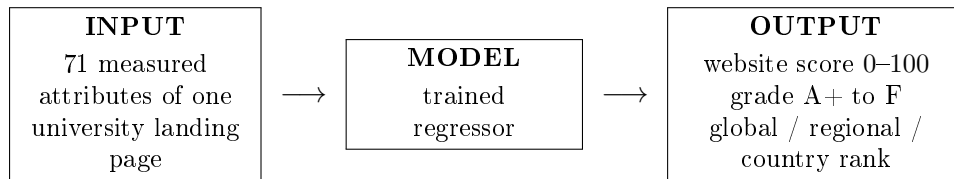
1 Introduction

1.1 Project Scope

For a prospective student, a university’s website is often the only source of information available before applying. It decides whether they can find the programmes on offer, learn the admission requirements, discover a deadline, or contact someone. A badly organised site is not merely an aesthetic failing; it is a barrier to access.

Most existing tools for assessing university website quality are rule-based: an analyst fixes a set of weights and the tool applies arithmetic. This project builds a machine-learning framework instead. It extracts landing-page attributes automatically, converts them into a documented quality score, and trains a range of regression algorithms to predict that score from the raw attributes. We apply it to 1,225 university websites worldwide, then test it on one country, Bangladesh, whose universities are kept out of training by construction.

1.2 Problem Statement



The task is supervised regression: given the attribute vector of a landing page, predict its website quality score. Since the deliverable is a league table, we treat rank agreement (Spearman ρ) as a primary metric alongside R^2 and mean absolute error.

1.3 Objectives

- O1.** Collect and clean a global dataset of university landing-page attributes, and classify every attribute by page region, measurement type and scoring role.
- O2.** Construct a transparent, reproducible label that turns those attributes into a 0–100 quality score, stated as equations, using every attribute for which a reason to use it can be given.
- O3.** Split the data 80/20, holding an entire country out of training, and train a broad range of regression algorithms on identical inputs.
- O4.** Compare their performance and explain why they differ.
- O5.** Evaluate the framework on 22 Bangladeshi universities the model has never seen, and characterise where those websites gain and lose points.
- O6.** Produce global, regional and country rankings, and measure how much of the ranking is a property of the websites rather than of our own weighting judgement.

Deliverables: the cleaned and scored dataset, the attribute catalogue, an executed analysis notebook, model artefacts with a scoring entry point, the ranking tables, and this report.

1.4 Contributions

1. A seven-dimension scoring model with a documented derivation, explicit equations, published inner weights, two gating rules and six non-linear response curves (Section 5).
2. A classification of all 71 attributes on three axes, specified in Table 3, with a stated reason for each of the 20 attributes the label does not use (Section 5.7).
3. A controlled comparison of twelve regression algorithms across six model families (Section 7).
4. A country-holdout case study in which all 22 Bangladeshi universities are excluded from training and scored as unseen data (Section 9).
5. An error analysis tracing the performance gap between model families to a specific mechanism rather than to capacity in the abstract (Section 8.1).

6. A sensitivity and ablation study measuring how much of the ranking survives a different analyst’s weights, and which blocks of attributes the model needs (Sections 8.3–8.4).

2 Literature Review

Work on university website quality has used automated web analysis, questionnaire studies, usability assessment and web-quality models. Table 1 summarises the studies closest to ours, with attention to those evaluating sites through automatically measurable content, usability, accessibility, performance and technical attributes.

Table 1: Summary of the literature most relevant to this framework.

#	Reference	Focus / method	What we reuse
1	Rashida <i>et al.</i> (2021), <i>Computers</i> 10(5):57	A multi-method framework for evaluating university websites in Bangladesh. The automated tool crawls with Selenium and evaluates three criteria: content of information (25 key strings), website performance and loading time (20 browser timing variables). A questionnaire study compares the survey ranking with the automated one.	The principal baseline. Most of our content attributes — vision and mission, faculty, notice board, library, contact — originate in or extend its content criteria. We extend its automated evaluation from a national to a global corpus, and replace its fixed weighting with a documented, sensitivity-tested one.
2	Saleh <i>et al.</i> (2022), <i>Indonesian J. of Electrical Engineering and Computer Science</i> 25(1):511–520	A systematic literature review of university website quality evaluation. It extracts 79 quality factors from prior work and consolidates them into six: information quality, specific content, usability, web appearance, service interaction quality and functionality. It reports the absence of any universally accepted model.	The six-factor structure is the theoretical foundation for grouping and justifying our attribute blocks. Its finding that no accepted model exists is what obliges us to construct a label and to state its construction in full (Section 5).
3	Mukanda, Mbuguah & Wabwoba (2022), <i>Int. J. of Computer Trends and Technology</i> 70(2):1–9	Evaluates the usability of the top five Kenyan university websites using automated tools (Nibbler, Silk-tide, Semrush, Ahrefs), covering responsiveness, colour contrast, broken links, speed, compression, SEO, and accessibility.	The source of our technical and accessibility attributes: mobile responsiveness, contrast ratio, broken links, load speed, compression, SEO metadata and alt-text coverage. Where it evaluates five universities, we evaluate 1,225.
4	Biyyapu <i>et al.</i> (2023), <i>Computers</i> 12(9):181	Applies machine learning to website quality classification from automatically extracted completeness attributes (missing URLs, images, videos, PDFs, form fields), comparing an MLP against SVM and other classifiers.	Motivates the move from a fixed rule-based score to a learned model, and the practice of comparing several algorithms on one feature vector. Our feature space is substantially broader, and Section 8.1 goes beyond reporting which family wins to identifying the structural reason it wins.

Two points from this literature shaped our design. Saleh *et al.* [2] report that no universally accepted model of university website quality exists, so there is no external label to borrow and one has to be constructed and stated in full (Section 5). Rashida *et al.* [1] and Mukanda *et al.* [3] both aggregate measured attributes using weights the analyst fixes, and neither reports what the ranking would look like under different weights. We measure that directly in Section 8.4.

3 Data Collection

3.1 Population and Frame

The dataset was built from university websites in six geographical regions. We crawled 1,230 landing pages. The collection instrument specified 75 candidate attributes; six of them (a19, a52, a55, a56, a64, a68) could not be extracted reliably and are absent from the data. We did not impute them, leaving **69 attributes per university**.

Collection was distributed across six collectors, one per region. One consequence recurs throughout this report: region and collector are perfectly confounded, so any regional difference in a measured quantity is partly a property of the websites and partly a property of who measured them.

3.2 Landing-Page Attributes Used for Evaluation

The attributes are limited to information visible on, or directly verifiable from, the landing page. Most are Boolean values recording the presence or absence of a page element, which is scraper-friendly and allows reliable DOM- and keyword-based detection. Integer attributes record counts such as navigation items, events, accreditation badges and broken links. Float attributes record continuous measurements such as page-load speed, colour contrast ratio, mobile responsiveness and alt-text coverage.

Table 3 is the full specification. Attributes are grouped by the block of Section 4.1. *Dim* names the dimension of Section 5 that consumes the attribute, or a dash where it does not enter the label, in which case the reason appears in the justification column; *Pts* is $w_k \lambda_{k,j}$, the points the attribute contributes at its maximum. The provenance keys record where each attribute comes from, with those marked *New* added from direct observation of live landing pages rather than taken from prior work.

Table 2: Provenance keys used in the attribute catalogue.

Key	Provenance
P1	Mukanda, Mbuguah & Wabwoba (2022) — technical and usability criteria
P2	Rashida <i>et al.</i> (2021) — landing-page key strings and questionnaire
P3	Saleh <i>et al.</i> (2022) — six-factor systematic review structure
New	Added by this project from observation of live landing pages
Derived	Engineered in <code>build_dataset.py</code> ; see Section 4.2

Table 3: Landing-page attributes used for university website evaluation. Attribute numbers, names, data types, value domains and justifications are as specified for the collection instrument; the dataset column name is printed beneath each attribute name. *Dim* is the scoring dimension of Section 5 that consumes the attribute and *Pts* the points it contributes at its maximum; a dash marks an attribute that is observed and supplied to every model but does not enter the label.

#	Attribute	Type	Domain	Justification	Dim	Pts
B1. Header & navigation — Wayfinding and structural usability (6 attributes, 6 scored)						
1	Logo / institutional branding a01_logo	Boolean	{0,1}	Institutional identity	D7	0.18
2	Primary navigation menu presence a02_primary_nav	Boolean	{0,1}	Core wayfinding; a page without one cannot be navigated at all	D4	3.75
3	Number of top-level navigation items a03_nav_item_count	Integer	[0-20]	Navigation complexity: too few leads nowhere, too many is a wall of links	D4	3.75
4	Search bar a04_search_bar	Boolean	{0,1}	Fast task completion	D4	3.00

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#	Attribute	Type	Domain	Justification	Dim	Pts
5	Language toggle (native / English) <code>a05_language_toggle</code>	Boolean	{0,1}	Multilingual accessibility	D5	0.90
6	Breadcrumb / current-location cue <code>a06_breadcrumb</code>	Boolean	{0,1}	Recognition over recall	D4	1.50
B2. Rankings & recognition — Externally conferred prestige signals (7 attributes, 2 scored)						
7	QS World Ranking badge / mention <code>a07_qs_badge</code>	Boolean	{0,1}	Global prestige signal displayed on the page (<i>excluded: prestige display; excluded to prevent prestige leakage</i>)	–	–
9	National / local ranking mention <code>a09_national_rank</code>	Boolean	{0,1}	Local prestige signal (<i>excluded: prestige display; excluded to prevent prestige leakage</i>)	–	–
11	Accreditation badges <code>a11_accreditation</code>	Boolean	{0,1}	Credibility signal	D7	0.33
12	Accreditation badge count <code>a12_accred_count</code>	Integer	[0-10]	Credibility depth (<i>excluded: count depth of a badge already scored by presence</i>)	–	–
13	Recent achievements / awards section <code>a13_achievements</code>	Boolean	{0,1}	Institutional reputation	D7	0.24
14	Impact / highlight statistics block <code>a14_stats_block</code>	Boolean	{0,1}	Quantified credibility (<i>excluded: marketing block; serves no applicant information need</i>)	–	–
15	Number of highlight statistics shown <code>a15_stat_item_count</code>	Integer	[0-10]	Richness of proof points (<i>excluded: count depth of a marketing block</i>)	–	–
B3. Notices & updates — Content currency and maintenance (6 attributes, 6 scored)						
16	Notice board presence <code>a16_notice_board</code>	Boolean	{0,1}	Core recurring information need	D3	1.20
17	Notice timestamp shown <code>a17_notice_timestamp</code>	Boolean	{0,1}	Signals content freshness	D3	0.90
18	Days since the most recent dated notice <code>notice_recency_days</code>	Derived	[0-3650]	Turns attribute 18 (an ISO date) into a staleness measure comparable across a crawl that spans time	D3	4.80
20	News & events section <code>a20_news_events</code>	Boolean	{0,1}	Institutional activity visibility	D3	2.40
21	Events / academic calendar link <code>a21_calendar_link</code>	Boolean	{0,1}	Time-sensitive planning information	D3	1.80
22	Admission announcements / deadlines <code>a22_admission_notice</code>	Boolean	{0,1}	High-frequency prospective-student query	D2	4.40
B4. Events & media — Engagement and multimedia richness (9 attributes, 3 scored)						
23	Upcoming events section <code>a23_upcoming_events</code>	Boolean	{0,1}	Engagement and currency	D3	0.60
24	Number of upcoming events listed <code>a24_event_count</code>	Integer	[0-20]	Activity richness	D3	3.00
25	Event images present <code>a25_event_images</code>	Boolean	{0,1}	Visual engagement (<i>excluded: media richness; presence is measurable, quality is not</i>)	–	–
26	Event captions / descriptions present <code>a26_event_captions</code>	Boolean	{0,1}	Informativeness (<i>excluded: media richness; presence is measurable, quality is not</i>)	–	–
27	Event date & time displayed <code>a27_event_datetime</code>	Boolean	{0,1}	Actionability: an undated event cannot be attended	D3	<i>mod.</i>
28	Forthcoming contests or competitions <code>a28_contests</code>	Boolean	{0,1}	Student engagement signal (<i>excluded: engagement signal; serves no applicant information need</i>)	–	–

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#	Attribute	Type	Domain	Justification	Dim	Pts
29	Video content on landing page a29_video_content	Boolean	{0,1}	Rich media engagement (<i>excluded: media richness; presence is measurable, quality is not</i>)	–	–
30	Photo / image gallery a30_image_gallery	Boolean	{0,1}	Visual storytelling (<i>excluded: media richness; presence is measurable, quality is not</i>)	–	–
31	Social media feed embed a31_social_feed_embed	Boolean	{0,1}	Real-time engagement (<i>excluded: media richness; present on 5.6% of sites</i>)	–	–
B5. Landing-page content — Information completeness (15 attributes, 15 scored)						
32	Vision & mission statement a32_vision_mission	Boolean	{0,1}	Institutional identity signal	D7	0.60
33	About / overview blurb a33_about_blurb	Boolean	{0,1}	First-impression orientation	D7	0.42
34	Department list / quick links a34_department_links	Boolean	{0,1}	Primary navigation need for an applicant	D1	7.00
35	Faculty highlight or link a35_faculty_link	Boolean	{0,1}	Academic credibility cue	D1	4.20
36	Research / publication highlight a36_research_highlight	Boolean	{0,1}	Academic reputation signal	D1	2.80
37	Academic programmes listing a37_programs_listing	Boolean	{0,1}	The decision-critical information a prospective student came for	D1	8.40
38	Scholarship information / link a38_scholarship	Boolean	{0,1}	Strong student interest area (53% in the P2 survey)	D2	3.96
39	Library access / link a39_library_link	Boolean	{0,1}	Common student resource	D1	3.36
40	Career / job portal link a40_career_link	Boolean	{0,1}	Student utility feature	D1	2.24
41	Alumni section / link a41_alumni_link	Boolean	{0,1}	Institutional trust signal	D7	0.30
42	FAQ link a42_faq_link	Boolean	{0,1}	Self-service reduces friction	D2	1.32
43	Contact information / web-mail link a43_contact_link	Boolean	{0,1}	Direct communication channel	D2	3.30
44	Student portal link a44_student_portal	Boolean	{0,1}	Core functional access	D7	0.33
45	Prospectus / brochure download a45_prospectus	Boolean	{0,1}	Decision-support material	D2	2.64
46	Admissions policy / procedure link a46_admissions_policy	Boolean	{0,1}	Decision-critical information	D2	5.72
B6. Footer details — Secondary navigation and credibility (5 attributes, 5 scored)						
47	Footer contact details a47_footer_contact	Boolean	{0,1}	Standard usability expectation	D7	0.42
48	Footer sitemap link a48_footer_sitemap	Boolean	{0,1}	Redundant navigation path	D4	1.50
49	Copyright / last-updated line a49_copyright_line	Boolean	{0,1}	Currency and credibility cue	D3	0.30
50	Social media links a50_social_links	Boolean	{0,1}	Engagement / communication channel	D7	0.12
51	Quick-links block a51_quick_links	Boolean	{0,1}	Reduces clicks to key pages	D4	1.50
B7. Visual design — Appearance and readability (3 attributes, 1 scored)						
53	Colour contrast ratio a53_contrast_ratio	Float	[1-21] WCAG scale	Readability and accessibility	D5	2.30
54	Banner / carousel content a54_banner_carousel	Boolean	{0,1}	First-impression engagement (<i>excluded: visual presentation the dataset cannot assess for quality</i>)	–	–

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#	Attribute	Type	Domain	Justification	Dim	Pts
57	Prominent logo placement a57_logo_prominence	Boolean	{0,1}	Brand recall (<i>excluded: visual presentation the dataset cannot assess for quality</i>)	–	–
B8. Service interaction — Two-way interaction and trust (4 attributes, 2 scored)						
58	Live chat / chatbot presence a58_live_chat	Boolean	{0,1}	Real-time support	D2	0.66
59	Feedback / contact form on landing page a59_feedback_form	Boolean	{0,1}	Two-way interaction (<i>excluded: present on 1.9% of sites; too rare to carry weight</i>)	–	–
60	Trust / security seal displayed a60_trust_seal	Boolean	{0,1}	Perceived trust	D7	0.06
61	Testimonials / student stories section a61_testimonials	Boolean	{0,1}	Sense of community (<i>excluded: marketing; serves no applicant information need</i>)	–	–
B9. Technical performance — Speed, security, reliability (6 attributes, 5 scored)						
62	Page load speed, standardised within region load_time_z_region	Derived	z-score	Raw load time is confounded with the collector, since each of six collectors covered exactly one region; only the within-region form is used	D6	1.40
63	Mobile responsiveness score a63_mobile_score	Float	[0-100]	Multi-device access need	D5	3.20
65	HTTPS / SSL presence a65_https	Boolean	{0,1}	Security and user trust; an unencrypted site should not receive an application form	D6	1.82
66	Broken links, log-compressed broken_links_log	Derived	log(1+n)	The raw count is heavily right-skewed; the log keeps 0 vs 3 from being swamped by 40 vs 200 (<i>excluded: redundant transform of a66, which is scored</i>)	–	–
66	Broken links on landing page a66_broken_links	Integer	[0-N]	Reliability and credibility	D6	1.68
67	Page compression enabled (Gzip etc.) a67_gzip	Boolean	{0,1}	Faster, lighter load	D6	0.49
B10. SEO & metadata — Discoverability (3 attributes, 3 scored)						
69	Page title & meta description present a69_title_meta	Boolean	{0,1}	Search-engine visibility	D6	0.84
70	Favicon presence a70_favicon	Boolean	{0,1}	Brand recognition in-browser	D6	0.28
71	Sitemap / robots.txt reference a71_sitemap_robots	Boolean	{0,1}	Crawlability and indexing	D6	0.49
B11. Accessibility — Inclusive access (4 attributes, 3 scored)						
72	Alt-text coverage on images a72_alt_text_pct	Float	[0-100] %	Screen-reader accessibility	D5	1.80
73	Accessible design across devices a73_accessible_design	Boolean	{0,1}	Inclusive access for all	D5	1.00
74	Text-size / contrast accessibility toggle a74_a11y_toggle	Boolean	{0,1}	Inclusive design option	D5	0.80
75	Bookmark / save-page facility a75_bookmark	Boolean	{0,1}	Convenience functionality (<i>excluded: present on 1.0% of sites; a browser feature, not a page property</i>)	–	–
B12. Measurement quality — Whether the attribute could be measured at all (3 attributes, 0 scored)						
18	Notice date was never found notice_recency_days_was_missing	Derived	{0,1}	Records that attribute 18 could not be measured, so 'never measured' stays distinguishable from 'measured as stale' (<i>excluded: measurement metadata, not a property of the website</i>)	–	–

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#	Attribute	Type	Domain	Justification	Dim	Pts
53	Contrast ratio was never measured a53_contrast_ratio_was_missing	Derived	{0,1}	As above, for attribute 53 (<i>excluded: measurement metadata, not a property of the website</i>)	–	–
72	Alt-text coverage was never measured a72_alt_text_pct_was_missing	Derived	{0,1}	As above, for attribute 72 (<i>excluded: measurement metadata, not a property of the website</i>)	–	–

3.3 Attribute Selection and Collection Protocol

Ranking attributes such as QS World Ranking and Webometrics were recorded as nullable, since not every university displays them, and the most recent notice date was stored in ISO 8601 format where available. Two categories, *Rankings & Recognition* and *Events & Media*, were added to the schema after inspecting real landing pages rather than taken from prior work.

For each university we used the verified official domain as the starting URL. Boolean attributes were encoded as

$$x_i = \begin{cases} 1, & \text{if the attribute is detected,} \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

counts as integers, continuous measurements as floating-point values, and unavailable nullable measurements as null. The scraper read HTML and DOM elements, hyperlinks, metadata, text content, images, media elements, accessibility properties and technical response information. Pages needing JavaScript were rendered in a browser environment before extraction. Requests respected `robots.txt`; the crawler submitted no forms and took no action that modifies a target site.

3.4 Data Cleaning and Audit

Every step below is implemented in `src/build_dataset.py` and is reproducible from the raw file. Nothing was corrected silently.

Table 4: Data preparation log.

Issue	Scale	Action taken
Constant or empty columns	3 columns	Dropped; they carry no information
Exact duplicate column	a62	≡ One copy dropped
	load_time_s	
External ranking <i>values</i>	95–99% missing	Dropped: unusable, and they describe the institution rather than the website. The ranking <i>badge</i> flags are kept, since displaying a badge is a property of the page
Duplicate sites	10 rows → 5	Deduplicated on the <i>resolved</i> URL, so two entry points to one page count once
Notice dates after the crawl date	284 rows	Censored to the crawl date rather than deleted
Regional labels in the country field	169 rows	Flagged; country rankings exclude them
Final dataset	1,225 universities × 71 features	

Dropping the external ranking values is the one place where we discarded a predictor that would have helped. QS and Webometrics ranks correlate with our score, but they measure institutional prestige rather than the website; keeping them would let a model reason that a famous university must have a good website, which is the inference this project exists to avoid. Section 9.3 shows what that inference would have cost.

3.5 Missing and Nullable Values

Three attributes are missing for some universities, and the usual remedy of replacing them with the column median is wrong for all three. The median of `notice_recency_days` is one day, so median-filling would record the 468 universities with no dated notice anywhere on their site as having posted something yesterday, handing the strongest freshness signal in the dataset to the institutions that earned it least. Because absence has a known direction in each case, we fill at the worst defensible value and compute a `*_was_missing` indicator *before* the fill, so that “never measured” stays recoverable.

Table 5: Missing-value policy.

Attribute	Rows	Filled with	What absence means
<code>notice_recency_days</code>	468	3650 days	no dated notice exists anywhere
<code>a72_alt_text_pct</code>	4	0%	no image carries a text alternative
<code>a53_contrast_ratio</code>	1	1.0:1	no readable text block was found

4 Data Schema

4.1 Final Dataset Structure and Feature Blocks

After cleaning and feature engineering each university is one feature vector, and the modelling dataset is

$$X \in \mathbb{R}^{1,225 \times 71}, \quad y \in [0, 100]^{1,225}, \quad (2)$$

alongside identification fields, the banded grade and three rank outputs. The 71 features fall into eleven thematic blocks corresponding to regions of a landing page, plus a twelfth holding the measurement-failure indicators. Every block except B12 contributes to the label; B12 records whether an attribute could be measured at all, which is a property of our extraction rather than of the website.

Table 6: Feature blocks. *Dimensions* names the scoring dimensions of Section 5 that draw on each block.

#	Feature block	Attrs	Nature of the signal	Dimensions
B1	Header & navigation	6	Wayfinding and structural usability	<i>D4, D5, D7</i>
B2	Rankings & recognition	7	Externally conferred prestige signals	<i>D7</i>
B3	Notices & updates	6	Content currency and maintenance	<i>D2, D3</i>
B4	Events & media	9	Engagement and multimedia richness	<i>D3</i>
B5	Landing-page content	15	Information completeness	<i>D1, D2, D7</i>
B6	Footer details	5	Secondary navigation and credibility	<i>D3, D4, D7</i>
B7	Visual design	3	Appearance and readability	<i>D5</i>
B8	Service interaction	4	Two-way interaction and trust	<i>D2, D7</i>
B9	Technical performance	6	Speed, security, reliability	<i>D5, D6</i>
B10	SEO & metadata	3	Discoverability	<i>D6</i>
B11	Accessibility	4	Inclusive access	<i>D5</i>
B12	Measurement quality	3	Whether the attribute could be measured at all	—
Total		71		

Table 7: Features by measurement type. The dataset is dominated by binary presence flags, which is why Euclidean distance, and therefore k -NN, performs poorly on it (Section 7).

Measurement type	Features
binary flag	57
count	6
indicator (0/1)	3
log1p count	1
percentage 0-100	1
index 0-100	1
ratio	1
z-score (within region)	1
Total	71

4.2 Feature Engineering

Four derived features are added, each for a stated reason.

Table 8: Engineered features.

Derived feature	Why it exists
notice_recency_days	Converts a notice date (attribute 18) into a staleness measure in days relative to the crawl date. A date is not comparable across a crawl that spans time; a gap is.
load_time_z_region	Standardises load time within region. Raw load time is confounded with the collector: each of six collectors covered one region, and regional medians span 8.1 s to 12.9 s, so part of the difference is measurement rather than website. Raw <code>load_time_s</code> is withheld from the model.
broken_links_log	$\log(1+n)$ of a heavily right-skewed count, so the difference between 0 and 3 broken links is not swamped by the difference between 40 and 200.
Three *_was_missing indicators	Record measurement failure as a fact in its own right, computed before imputation.

4.3 Modelling Representation

Three groups of columns are withheld from the model. Identification and crawl metadata, namely `uni_id`, `name`, `url`, `final_url`, `http_status`, `fetches_at` and `member`, are not properties of the website. Raw `load_time_s` is replaced by its within-region standardisation. Most importantly, the intermediate outputs of the scoring curves (`nav_quality`, `notice_evidence`, `event_evidence`, `load_time_pct_region`) are withheld, because supplying them would hand the model the answer. It receives the raw observables instead, such as `a03_nav_item_count` and `a24_event_count`, and has to recover the threshold, saturation and gate behaviour on its own.

All 71 remaining features go to every model, including the 20 that do not enter the label, so the model must separate an attribute that matters from one that merely accompanies mattering. Section 8.3 shows that it does.

4.4 Exploratory Analysis

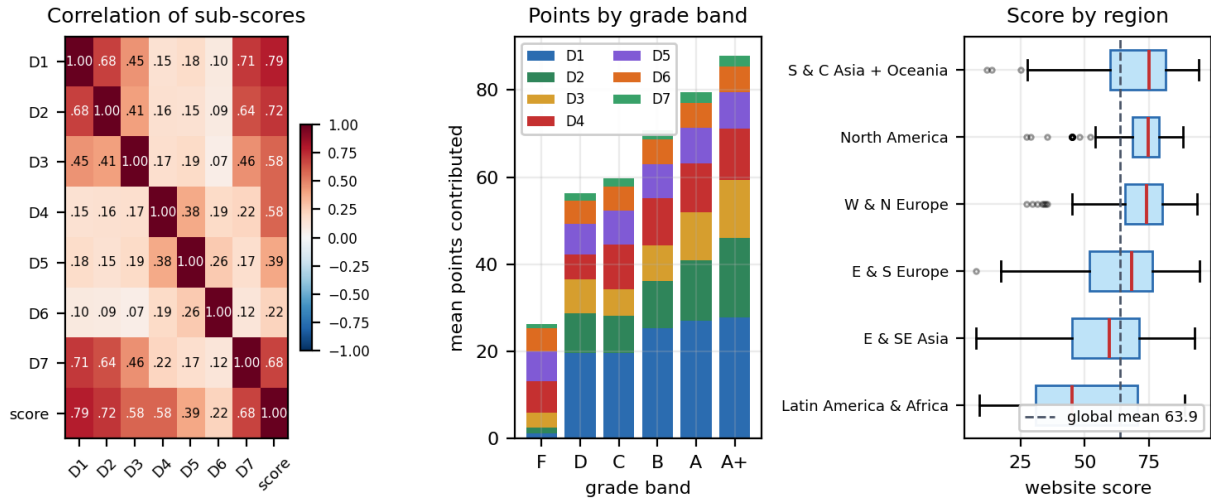


Figure 1: Left: correlation between the seven sub-scores and the final score. Centre: mean points contributed by each dimension, by grade band. Right: score distribution by region.

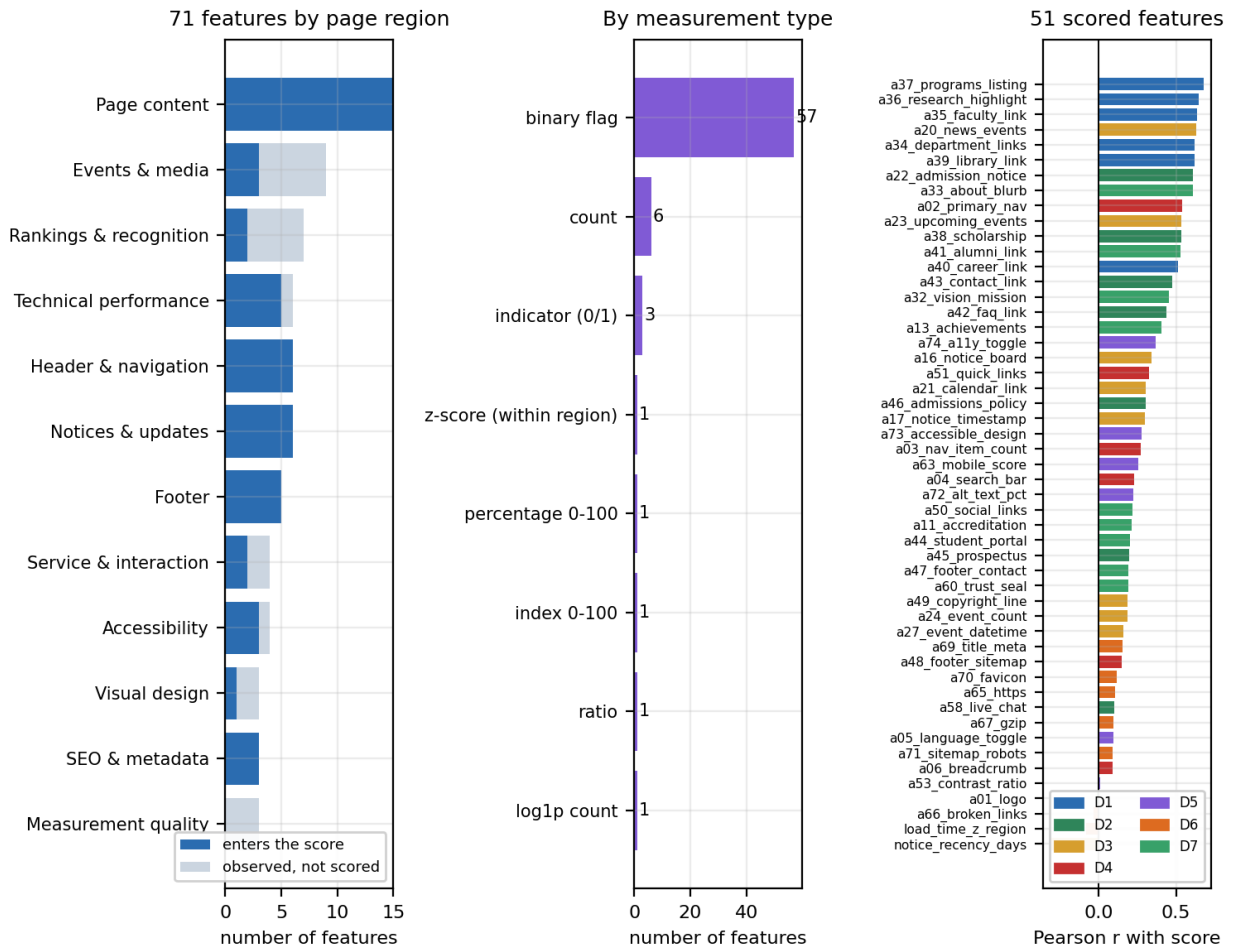


Figure 2: Left: features by the page region they came from, split by whether they enter the label. Centre: by measurement type. Right: the 51 scored features and their raw correlation with the score, coloured by dimension.

Three properties of the data shape the modelling that follows. The dimensions are only loosely related to each other: a university that publishes its programmes well is not reliably the one that keeps a fresh notice board, which is why a multi-dimensional score is worth computing rather than collapsing into one number. They also differ sharply in how far they separate universities (Table 9); nearly every site does equally well on technical quality, since 98.9% use HTTPS, so that dimension cannot separate anybody whatever weight it is given. This is visible before any model is fitted and explains most of the gap between declared weight and realised importance in Section 8.2. Finally, the score distribution is bimodal rather than bell-shaped: the shoulder near 45 in Figure 4 is the population of sites hitting a scoring gate, and it is what separates the algorithm families in Section 8.1.

Table 9: Spread of each dimension across the 1,225 universities. A dimension can differentiate only to the extent that universities actually differ on it.

	D_1	D_2	D_3	D_4	D_5	D_6	D_7
Weight	28	22	15	15	10	7	3
Mean	0.785	0.483	0.568	0.643	0.773	0.799	0.652
Std. dev.	0.318	0.241	0.269	0.222	0.131	0.115	0.215
Inter-quartile range	0.270	0.240	0.460	0.225	0.159	0.157	0.250

5 Label Construction

5.1 The Need for a Constructed Label

The dataset records what a website has, not how good it is. There is no column called “quality”, and as the review literature reports, no accepted model to borrow one from. Published university rankings measure research output and reputation rather than websites, so using them would mean training a model to predict prestige from page structure.

A label therefore has to be constructed. It is the most consequential design decision here and the one most open to disagreement, so this section states how the dimensions were derived, how the weights were set, what the equations are, which attributes are deliberately left out, and what the known weaknesses of the result are.

5.2 Derivation of the Seven Dimensions

We did not obtain the dimensions by clustering the attributes or by factor analysis. They come from a task decomposition driven by one question: what does a prospective student need from a university website? The applicant’s path through a landing page breaks into seven sub-tasks; each becomes a dimension, and the attributes serving it become its terms.

Table 10: Derivation of the seven dimensions from the applicant’s task.

	The applicant’s sub-task	Attributes that serve it	Blocks	Dim.
1	“What can I study here, and who will teach me?”	programmes listing, department links, faculty page, research highlight, library, careers	B5	D_1
2	“How do I apply, what does it cost, and who do I ask?”	admissions policy, admission notices, scholarships, prospectus, contact page, FAQ, live chat	B3, B5, B8	D_2
3	“Is this institution actually alive and running?”	notice recency, notice board and timestamp, dated events, news, calendar, last-updated line	B3, B4, B6	D_3
4	“Can I find any of the above?”	primary navigation, menu breadth, search box, breadcrumb, quick links, footer sitemap	B1, B6	D_4
5	“Can I read and use this, on the phone I own?”	mobile score, contrast ratio, alt-text coverage, accessible design, language toggle, accessibility toggle	B1, B7, B9, B11	D_5
6	“Does the site work, can it be found, and is it safe?”	HTTPS, broken links, load speed, compression, page title and meta, sitemap, favicon	B9, B10	D_6
7	“Is it clear who these people are?”	vision/mission, about text, footer contact, accreditation, student portal, alumni, achievements, logo, social links, trust seal	B1, B2, B5, B6, B8	D_7

Three rules governed the assignment, and they are what make the mapping reproducible rather than a matter of taste. First, one attribute belongs to one dimension, assigned to the sub-task it primarily serves: `a48_footer_sitemap` sits in the footer block but serves finding things, so D_4 scores it and D_7 does not, since double-counting would quietly re-weight the label. Second, an attribute serving no sub-task is not scored, but the reason must be stated; Section 5.7 gives it for each of the 20 excluded attributes. Third, only attributes that are actually populated are used: anything missing for more than half the dataset was dropped in cleaning, and anything present on fewer than 2% of sites is excluded as too rare to carry weight.

5.3 Weight Assignment

The seven dimensions combine through weights summing to 100, set by a stated ordering principle rather than by tuning: a dimension’s weight is proportional to how directly it serves the applicant’s purpose in visiting the site. That yields three tiers.

Table 11: The three weighting tiers.

Tier	Principle	Dimensions	Points
The content	What the applicant came for. If it is absent, nothing else rescues the visit.	D_1, D_2	50
The delivery	The machinery that makes the content reachable, current and usable. Necessary, but in service of the content.	D_3, D_4, D_5	40
The foundation	Baseline hygiene, discoverability and self-presentation. Expected rather than rewarded; their absence is punished by the gates instead.	D_6, D_7	10

Within a tier the same principle sets the order. D_1 outranks D_2 because a student who cannot find

the programme has no reason to read the admission rules, and D_4 outranks D_3 and D_5 because navigation conditions access to everything else. These are not the only defensible weights, and we do not claim they are; we claim that a stated rule fixed them before any score was computed, that they are published in full, and that the league table barely moves when they are perturbed (Section 8.4).

5.4 The Scoring Equation

Each dimension is a sub-score $D_k \in [0, 1]$, and the website score is their weighted sum subject to a cap:

$$\boxed{\text{Score} = \min\left(\sum_{k=1}^7 w_k D_k, \text{cap}\right), \quad \sum_{k=1}^7 w_k = 100} \quad (3)$$

Each dimension is itself a weighted sum of transformed attributes,

$$D_k = \sum_j \lambda_{k,j} \phi_{k,j}(a_{k,j}), \quad \sum_j \lambda_{k,j} = 1, \quad (4)$$

where $a_{k,j}$ is a measured attribute, $\phi_{k,j}$ the transform applied to it (the identity for a binary flag, otherwise one of the response curves of Section 5.5), and $\lambda_{k,j}$ its weight inside the dimension. Since the inner weights sum to one, dimension k contributes at most w_k points and attribute j of dimension k is worth exactly $w_k \lambda_{k,j}$ points at its maximum. Writing $g(a)$ for an attribute read as 0/1:

$$D_1 = 0.30 g(\text{programs}) + 0.25 g(\text{departments}) + 0.15 g(\text{faculty}) \\ + 0.12 g(\text{library}) + 0.10 g(\text{research}) + 0.08 g(\text{careers}) \quad (5)$$

$$D_2 = 0.26 g(\text{admissions policy}) + 0.20 g(\text{admission notice}) + 0.18 g(\text{scholarship}) \\ + 0.15 g(\text{contact}) + 0.12 g(\text{prospectus}) + 0.06 g(\text{FAQ}) + 0.03 g(\text{live chat}) \quad (6)$$

$$D_3 = 0.32 f_{\text{rec}}(\text{notice age}) + 0.20 f_{\text{evt}}(\text{event count, dated}) + 0.16 g(\text{news}) \\ + 0.12 g(\text{calendar}) + 0.08 g(\text{notice board}) + 0.06 g(\text{timestamp}) \\ + 0.04 g(\text{upcoming events}) + 0.02 g(\text{last-updated line}) \quad (7)$$

$$D_4 = 0.25 g(\text{primary nav}) + 0.25 f_{\text{nav}}(\text{menu items}) + 0.20 g(\text{search}) \\ + 0.10 g(\text{breadcrumb}) + 0.10 g(\text{quick links}) + 0.10 g(\text{footer sitemap}) \quad (8)$$

$$D_5 = 0.32 \frac{\text{mobile score}}{100} + 0.23 f_{\text{con}}(\text{contrast}) + 0.18 f_{\text{alt}}(\text{alt-text \%}) \\ + 0.10 g(\text{accessible design}) + 0.09 g(\text{language toggle}) + 0.08 g(\text{a11y toggle}) \quad (9)$$

$$D_6 = 0.26 g(\text{https}) + 0.24 f_{\text{brk}}(\text{broken links}) + 0.20 (1 - \text{pctl}_{\text{region}}(\text{load time})) \\ + 0.12 g(\text{title \& meta}) + 0.07 g(\text{gzip}) + 0.07 g(\text{sitemap/robots}) + 0.04 g(\text{favicon}) \quad (10)$$

$$D_7 = 0.20 g(\text{vision}) + 0.14 g(\text{about}) + 0.14 g(\text{footer contact}) + 0.11 g(\text{accreditation}) \\ + 0.11 g(\text{student portal}) + 0.10 g(\text{alumni}) + 0.08 g(\text{achievements}) \\ + 0.06 g(\text{logo}) + 0.04 g(\text{social links}) + 0.02 g(\text{trust seal}) \quad (11)$$

Table 12 restates Equations (5)–(11) against the exact column names with the points each attribute carries. It is generated from `data/feature_catalog.csv`, so the equations printed here and the code that computes the score cannot drift apart.

Table 12: Every term of every dimension: the attribute, its weight $\lambda_{k,j}$ inside the dimension, the points it is worth at its maximum, and the transform $\phi_{k,j}$ applied before weighting. *mod.* marks `a27_event_datetime`, which is not an additive term but the multiplier inside f_{evt} .

Attribute	$\lambda_{k,j}$	Points	Transform
D1 — Academic information $w_1 = 28$ points			
a37_programs_listing	0.30	8.40	identity (0/1)
a34_department_links	0.25	7.00	identity (0/1)
a35_faculty_link	0.15	4.20	identity (0/1)
a39_library_link	0.12	3.36	identity (0/1)
a36_research_highlight	0.10	2.80	identity (0/1)
a40_career_link	0.08	2.24	identity (0/1)
<i>sum</i>	1.00	28.00	
D2 — Admission support $w_2 = 22$ points			
a46_admissions_policy	0.26	5.72	identity (0/1)
a22_admission_notice	0.20	4.40	identity (0/1)
a38_scholarship	0.18	3.96	identity (0/1)
a43_contact_link	0.15	3.30	identity (0/1)
a45_prospectus	0.12	2.64	identity (0/1)
a42_faq_link	0.06	1.32	identity (0/1)
a58_live_chat	0.03	0.66	identity (0/1)
<i>sum</i>	1.00	22.00	
D3 — Currency and activity $w_3 = 15$ points			
notice_recency_days	0.32	4.80	recency_curve, 6-step decay
a24_event_count	0.20	3.00	event_curve, 4-step x date-time factor
a20_news_events	0.16	2.40	identity (0/1)
a21_calendar_link	0.12	1.80	identity (0/1)
a16_notice_board	0.08	1.20	identity (0/1)
a17_notice_timestamp	0.06	0.90	identity (0/1)
a23_upcoming_events	0.04	0.60	identity (0/1)
a49_copyright_line	0.02	0.30	identity (0/1)
a27_event_datetime	<i>mod.</i>	<i>mod.</i>	modifier: x1.0 if events are dated, x0.6 if not
<i>sum</i>	1.00	15.00	
D4 — Navigation and findability $w_4 = 15$ points			
a03_nav_item_count	0.25	3.75	nav_breadth_curve, inverted U
a02_primary_nav	0.25	3.75	identity (0/1); also triggers the 45-point gate
a04_search_bar	0.20	3.00	identity (0/1)
a48_footer_sitemap	0.10	1.50	identity (0/1)
a51_quick_links	0.10	1.50	identity (0/1)
a06_breadcrumb	0.10	1.50	identity (0/1)
<i>sum</i>	1.00	15.00	
D5 — Usability and accessibility $w_5 = 10$ points			
a63_mobile_score	0.32	3.20	linear rescale 0-100 -> 0-1
a53_contrast_ratio	0.23	2.30	contrast_curve, plateau at 7:1
a72_alt_text_pct	0.18	1.80	sqrt of 0-100 rescale (concave)
a73_accessible_design	0.10	1.00	identity (0/1)
a05_language_toggle	0.09	0.90	identity (0/1)
a74_a11y_toggle	0.08	0.80	identity (0/1)
<i>sum</i>	1.00	10.00	
D6 — Technical quality and discoverability $w_6 = 7$ points			
a65_https	0.26	1.82	identity (0/1); also triggers the 60-point gate

continued on next page

Table 12, continued from previous page

Attribute	$\lambda_{k,j}$	Points	Transform
a66_broken_links	0.24	1.68	broken_curve, 5-step penalty
load_time_z_region	0.20	1.40	within-region percentile, inverted
a69_title_meta	0.12	0.84	identity (0/1)
a71_sitemap_robots	0.07	0.49	identity (0/1)
a67_gzip	0.07	0.49	identity (0/1)
a70_favicon	0.04	0.28	identity (0/1)
<i>sum</i>	1.00	7.00	
D7 — Institutional identity and transparency $w_7 = 3$ points			
a32_vision_mission	0.20	0.60	identity (0/1)
a47_footer_contact	0.14	0.42	identity (0/1)
a33_about_blurb	0.14	0.42	identity (0/1)
a11_accreditation	0.11	0.33	identity (0/1)
a44_student_portal	0.11	0.33	identity (0/1)
a41_alumni_link	0.10	0.30	identity (0/1)
a13_achievements	0.08	0.24	identity (0/1)
a01_logo	0.06	0.18	identity (0/1)
a50_social_links	0.04	0.12	identity (0/1)
a60_trust_seal	0.02	0.06	identity (0/1)
<i>sum</i>	1.00	3.00	
All seven dimensions		100	

5.5 Non-Linear Response Curves

Six attributes do not reward linearly, and the label says so explicitly rather than assuming proportionality. Each transform is chosen because the attribute’s relationship with quality is genuinely not monotone or not proportional.

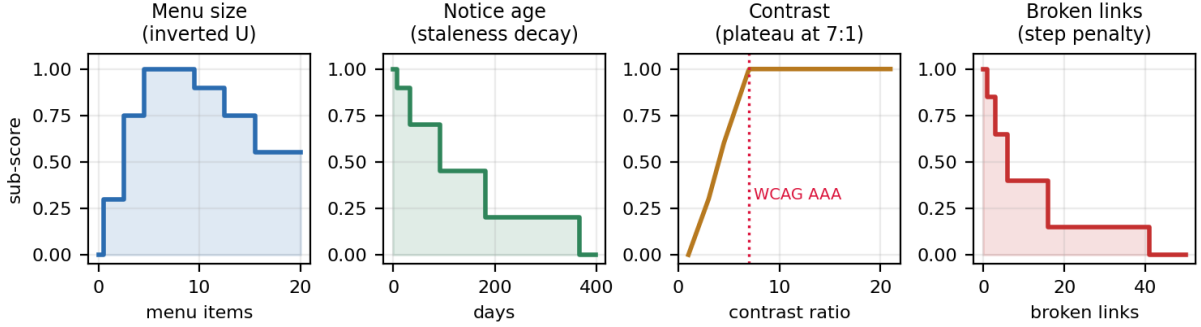


Figure 3: Four of the response curves. The remaining two are f_{evt} (Equation (16)) and the concave f_{alt} (Equation (17)).

Menu breadth, an inverted U. A menu of 0–2 items leads nowhere, 5–9 items indicates an organised site, and 20 items is a wall of links. Neither “more is better” nor “less is better” describes

this, so no monotone function can represent it. The curve enters D_4 :

$$f_{\text{nav}}(n) = \begin{cases} 0.00 & n \leq 0 \\ 0.30 & 1 \leq n \leq 2 \\ 0.75 & 3 \leq n \leq 4 \\ \mathbf{1.00} & \mathbf{5 \leq n \leq 9} \\ 0.90 & 10 \leq n \leq 12 \\ 0.75 & 13 \leq n \leq 15 \\ 0.55 & n \geq 16 \end{cases} \quad (12)$$

Notice age, a staleness decay. A post from last week and one from last year are not 51 weeks apart in value; “recent” and “abandoned” are plateaus with a cliff between them. This term carries the largest single weight in D_3 :

$$f_{\text{rec}}(d) = \begin{cases} 1.00 & d \leq 7 \text{ days} \\ 0.90 & d \leq 30 \\ 0.70 & d \leq 90 \\ 0.45 & d \leq 180 \\ 0.20 & d \leq 365 \\ 0.00 & d > 365 \end{cases} \quad (13)$$

Contrast, a plateau at WCAG AAA. The accessibility standard is 7:1, and beyond it more contrast does not improve reading, since 21:1 is no better than 12:1. The curve therefore stops rewarding instead of rising:

$$f_{\text{con}}(r) = \begin{cases} 0.30 u(r; 1, 3) & r < 3.0 \\ 0.30 + 0.30 u(r; 3, 4.5) & 3.0 \leq r < 4.5 \\ 0.60 + 0.40 u(r; 4.5, 7) & 4.5 \leq r < 7.0 \\ 1.00 & r \geq 7.0 \end{cases} \quad u(x; l, h) = \frac{x - l}{h - l} \Big|_0^1 \quad (14)$$

Broken links, a step penalty. One dead link is an oversight and forty is neglect, so the penalty is not proportional to the count:

$$f_{\text{brk}}(n) = \begin{cases} 1.00 & n = 0 \\ 0.85 & 1 \leq n \leq 2 \\ 0.65 & 3 \leq n \leq 5 \\ 0.40 & 6 \leq n \leq 15 \\ 0.15 & 16 \leq n \leq 40 \\ 0.00 & n > 40 \end{cases} \quad (15)$$

Events, a count scaled by whether they are dated. Twenty undated “events” form a decorative block rather than a calendar, so the count curve in D_3 is multiplied by a factor reflecting whether any of them carries a date:

$$f_{\text{evt}}(c, t) = \underbrace{\begin{cases} 0.00 & c = 0 \\ 0.55 & 1 \leq c \leq 2 \\ 1.00 & 3 \leq c \leq 12 \\ 0.85 & 13 \leq c < 20 \\ 0.70 & c \geq 20 \end{cases}}_{\text{how many}} \times \underbrace{\begin{cases} 1.0 & t = 1 \\ 0.6 & t = 0 \end{cases}}_{\text{are they dated?}} \quad (16)$$

Alt-text, concave. Going from 0% to 20% of images labelled is a larger accessibility gain than going from 80% to 100%, so the transform in D_5 is

$$f_{\text{alt}}(p) = \sqrt{p/100}, \quad p \in [0, 100]. \quad (17)$$

5.6 Gating Rules

Two failures are severe enough that no amount of other content compensates. We express them as hard caps rather than large negative weights, because that is what they mean.

Table 13: The two gating rules.

Condition	Cap	Affected	Rationale
no primary navigation	45	182	A site that cannot be navigated cannot serve an applicant, however much content is buried inside it
no HTTPS	60	14	A site that is not encrypted should not be trusted with an application form or personal data
<i>at least one gate</i>		186	15.2% of the dataset; 10 universities trigger both

The gates destroy 2,652 points in total, an average of 18.8 for each of the 186 universities they touch. That is the arithmetic behind the shoulder at 45 in Figure 4, it is what makes Equation (3) non-linear, and it turns out to account for most of the difference between the twelve algorithms compared later.

5.7 Attributes Deliberately Excluded from the Label

51 of the 71 features enter the label and 20 do not. Rule two of Section 5.2 requires a reason for each, and Table 14 gives them.

Table 14: The exclusion register. All 71 features, including these, are still supplied to every model as input; only the target omits them.

Reason for exclusion	Attrs	Attributes
Prestige leakage	2	a07_qs_badge, a09_national_rank
Quality not judgeable from presence	7	a25_event_images, a26_event_captions, a29_video_content, a30_image_gallery, a31_social_feed_embed, a54_banner_carousel, a57_logo_prominence
No applicant information need	3	a14_stats_block, a28_contests, a61_testimonials
Depth count of a scored presence	2	a12_accred_count, a15_stat_item_count
Too rare to carry weight	2	a59_feedback_form, a75_bookmark
Redundant with a scored attribute	1	broken_links_log
Measurement metadata	3	a53_contrast_ratio_was_missing, a72_alt_text_pct_was_missing, notice_recency_days_was_missing
Total excluded	20	of 71 features; the other 51 enter the label

Two categories deserve comment. *Quality not judgeable from presence* is the largest: the crawler can record that a page has an image gallery, a carousel or an embedded video, but not whether

any of them is any good, and scoring their presence would reward having a carousel rather than having a usable one. *Prestige leakage* keeps the two ranking-badge flags out of the label for the same reason the ranking value columns were dropped in cleaning, since a website-quality score that rewards displaying a QS badge is partly measuring institutional fame.

5.8 Worked Example and Resulting Label

The label is auditable end to end. `data/dimension_scores.csv` stores every sub-score, point contribution, cap and final score for all 1,225 universities, and the notebook checks that recomputing Equation (3) from those sub-scores reproduces the published score for 100% of them. One row in full:

Table 15: Khulna University of Engineering & Technology (`kuet.ac.bd`), scored step by step. No gate applies, so the cap is 100 and the score is the plain weighted sum. Small discrepancies against the published score are rounding, since the published value is computed from unrounded sub-scores.

	Dimension	Sub-score D_k	Weight w_k	Points
$D1$	Academic information	0.920	28	25.76
$D2$	Admission support	0.850	22	18.70
$D3$	Currency and activity	0.908	15	13.62
$D4$	Navigation and findability	0.600	15	9.00
$D5$	Usability and accessibility	0.749	10	7.49
$D6$	Technical quality and discoverability	0.707	7	4.95
$D7$	Institutional identity and transparency	0.890	3	2.67
Uncapped total $\sum_k w_k D_k$				82.19
Cap in force				100.00
Published score				82.18
Grade / global rank / national rank				A / 141 / 4

The diagnosis is immediate, and it is the kind of output the framework exists to produce: KUET’s weakest dimension by a wide margin is D_4 , which carries 15 points. Raising its navigation to the level of its own academic content would move it several points and roughly fifty global places.

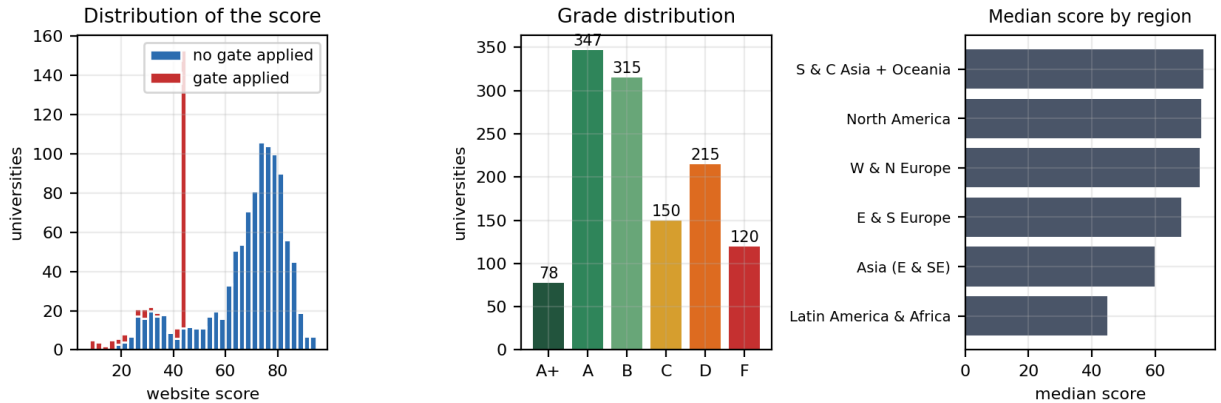


Figure 4: The website score across 1,225 universities. The shoulder near 45 in the left panel is the population to which a gating rule applies.

Table 16: Distribution of the label.

Mean	63.9	Grade A+ (≥ 85)	78
Standard deviation	18.5	Grade A (75–85)	347
Minimum	7.6	Grade B (65–75)	315
Maximum	94.9	Grade C (50–65)	150
Universities gated	186	Grade D (35–50)	215
		Grade F (< 35)	120

5.9 Threats to the Validity of the Label

The label is an arithmetic function of 51 of the 71 features the model is given. This is a serious limitation and the leading methodological risk in the study, and it bounds what the modelling results can be said to demonstrate, so we state it here rather than leave a reader to discover it.

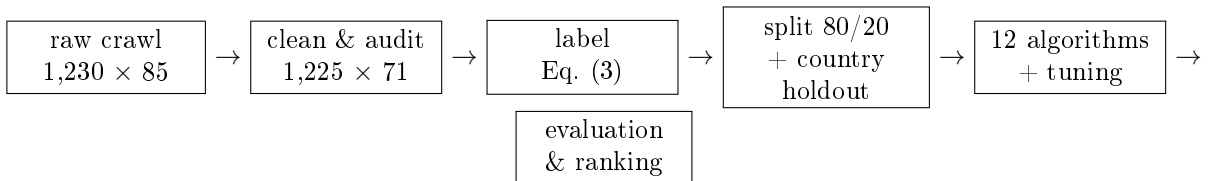
What the reported accuracy does and does not mean

Because y is computed from the features by Equation (3), a flexible enough model can in principle reconstruct it exactly. **The reported $R^2 = 0.9925$ is therefore not evidence that website quality is predictable**; it is evidence that this particular function is recoverable from the raw observables.

The question the report can legitimately answer is consequently narrower: given a scoring function containing thresholds, saturations and two hard gates, which families of regression algorithm can recover it from the raw attributes, and why do the others fail? That question is not circular, and Section 8.1 answers it with a mechanism rather than a leaderboard. Three properties of the design keep it meaningful. The model never sees the sub-scores, the curve outputs or the gate flag, only the raw attributes, so it has to recover the structure rather than read it off. 20 features do not enter the label at all, so it also has to work out what is irrelevant. And recovery is demonstrably not automatic: R^2 across the eleven non-trivial algorithms spans 0.768 to 0.990, with the spread concentrated on an identifiable subpopulation.

Two consequences follow, and Section 11 carries both. The ranking is a separate artefact from the model, and its trustworthiness rests on the defensibility of the weights rather than on R^2 , which is why Section 8.4 tests the ranking directly. And the gap between declared weight and realised importance in Section 8.2 is a statement about which dimensions vary between universities in this dataset, not a discovery about quality. The proper remedy is a label that is not a function of the features, either independent human ratings or blind pairwise judgements fitted with a Bradley–Terry model; that was outside the scope of this study and is the first item of future work.

6 ML Study — Pipeline Design



6.1 Train / Test Split and the Country Holdout

The dataset is split 80% training (981 universities) and 20% testing (244), stratified on score band so both halves span the full quality range, with a fixed random seed, and with one constraint imposed on top: **every one of the 22 Bangladeshi universities is in the test set, and none is used for training.**

There are two reasons. The first is that the case study has to mean one thing. Section 9 presents a table of all 22 Bangladeshi universities with predicted and actual scores; had some been in the training set, half the table would be a memory and half a prediction, and the two halves would not be comparable row to row. The second is that this is a stricter generalisation test than a random split. The 22 sites are not a random draw: they share a region, a hosting environment, a set of content-management templates and a set of design conventions. Scoring them after seeing none of them tests whether the model learned the scoring structure or the local habits of countries it had already met.

The rest of the test set is drawn band-stratified from the other 1,203 universities. We measured the cost of the constraint rather than assuming it: the largest distortion to any grade band is 1.2 percentage points. The test set is used once, in Section 7.2; no algorithm, hyper-parameter or threshold was chosen with it, and the build script asserts that no Bangladeshi university reaches the training set so that a later edit cannot break the holdout silently.

Table 17: Stratification after the country holdout. “Drift” is the test-set share of a band minus its share of the full dataset.

Score band	All	Train	Test	All %	Test %	Drift (pp)
F <35	120	98	22	9.8	9.0	-0.8
D 35-50	215	173	42	17.6	17.2	-0.4
C 50-65	151	122	29	12.3	11.9	-0.4
B 65-75	314	250	64	25.6	26.2	+0.6
A 75-85	347	275	72	28.3	29.5	+1.2
A+ >=85	78	63	15	6.4	6.1	-0.3
Total	1225	981	244	100.0	100.0	

6.2 Algorithms, Validation Protocol and Metrics

Twelve algorithms across six families were trained on identical data. Feature scaling is applied inside a pipeline for the distance-, kernel- and gradient-based methods that need it, and deliberately not for tree methods, which are invariant to monotone rescaling.

Table 18: The twelve algorithms, by family.

Family	Algorithms
Baseline	Mean predictor
Linear	Linear Regression, Ridge ($\alpha = 1$), Lasso ($\alpha = 0.1$)
Instance	k -Nearest Neighbours ($k = 5$)
Kernel	Support Vector Regression (RBF, $C = 100$)
Neural	Multi-Layer Perceptron (128×64 , $L_2 = 10^{-2}$, target-scaled)
Tree	Decision Tree
Ensemble	Random Forest, Extra Trees, Gradient Boosting, LightGBM

Model selection and tuning use 5-fold cross-validation on the training set only, with scalers fitted inside each fold so no information from a validation partition reaches the model. Hyper-parameters for the selected model come from a grid search over 81 combinations, again 5-fold on the training set. Two baselines bound the results: the mean predictor sets the floor, since any model failing to beat it has learned nothing, and linear regression sets the interpretable floor, being what an additive model of the kind the prior literature uses can reach.

We report R^2 for variance explained, MAE in score points because it is directly interpretable, RMSE because it penalises large errors, and Spearman ρ for rank agreement, which matters most given that the deliverable is a league table. For the case study we add Kendall τ and the number of national positions predicted exactly.

7 Results

7.1 Model Comparison and Cross-Validation

Table 19: Twelve algorithms on the held-out test set, ordered by R^2 . Cross-validation figures are 5-fold on the training set only.

	Model	Family	R^2	MAE	RMSE	Spearman	CV R^2
1	LightGBM	ensemble	0.990	1.31	1.80	0.989	0.988 ± 0.002
2	Gradient Boosting	ensemble	0.988	1.54	1.97	0.988	0.986 ± 0.000
3	Neural Net (MLP)	neural	0.976	1.56	2.74	0.987	0.967 ± 0.032
4	Random Forest	ensemble	0.974	2.10	2.86	0.971	0.965 ± 0.006
5	Extra Trees	ensemble	0.947	2.99	4.11	0.946	0.926 ± 0.012
6	Lasso	linear	0.945	2.86	4.18	0.984	0.948 ± 0.010
7	SVR (RBF kernel)	kernel	0.942	2.40	4.31	0.973	0.957 ± 0.016
8	Linear Regression	linear	0.933	2.86	4.61	0.983	0.949 ± 0.010
9	Ridge	linear	0.930	2.91	4.71	0.982	0.948 ± 0.011
10	Decision Tree	tree	0.929	3.38	4.75	0.939	0.901 ± 0.007
11	k-NN (k=5)	instance	0.768	5.37	8.60	0.868	0.798 ± 0.040
12	Mean baseline	baseline	-0.002	14.97	17.86	—	-0.009 ± 0.005

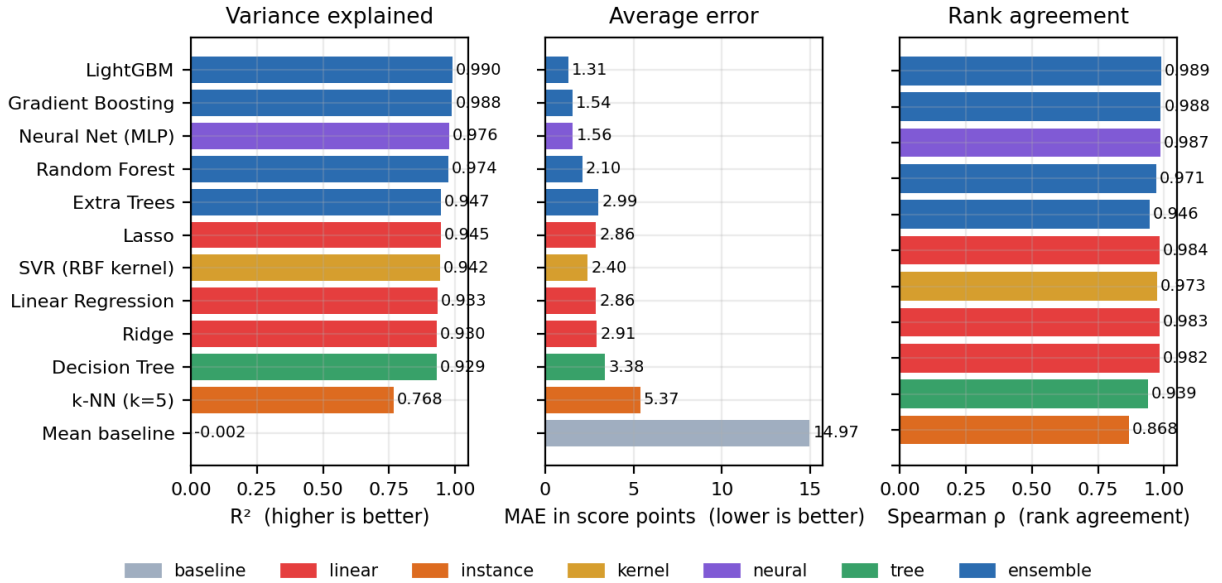


Figure 5: Model comparison on the held-out test set, coloured by algorithm family.

Every algorithm beats the mean baseline comfortably, so the attributes carry real signal, and the gradient-boosted ensembles lead on all four metrics. The worst performer is k -NN, which the feature representation predicts before the model is run: 57 of the 71 features are binary flags, and in a 71-dimensional, largely binary space Euclidean distance is a poor measure of similarity.

A single split can flatter or punish a model by chance, so we re-checked the ordering with 5-fold cross-validation on the training set alone. The two orderings agree at Spearman $\rho = 0.909$, so the comparison is not an artefact of one particular split. One practical note: the MLP was the only model that needed the target scaled as well as the inputs, since on the raw 0–100 scale one fold in five diverged and cross-validated R^2 collapsed. Target scaling and L_2 regularisation stabilised it, but its cross-validation standard deviation remains an order of magnitude larger than LightGBM’s, so it is not only slightly worse on average but markedly less reliable.

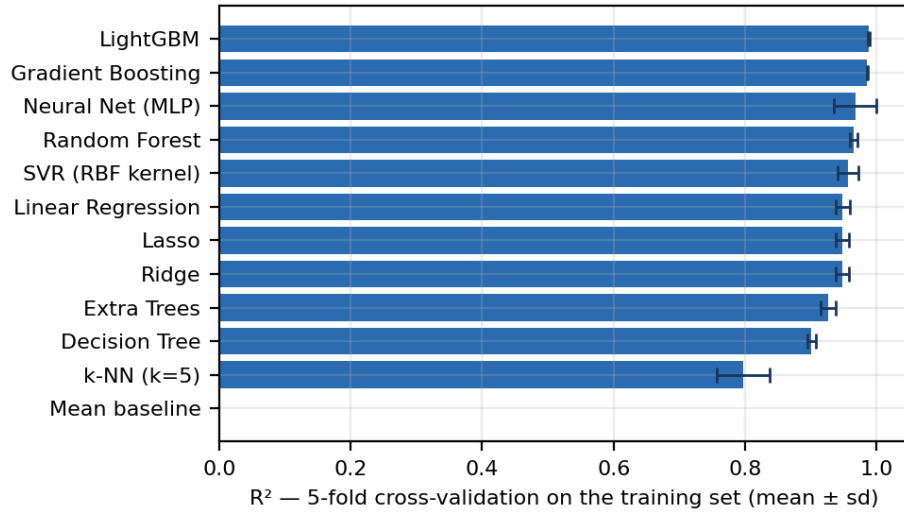


Figure 6: Cross-validated R^2 (mean $\pm$ standard deviation) on the training set.

7.2 Tuning and Final Held-Out Evaluation

Table 20: Grid searched; chosen values in bold.

Parameter	Values
n_estimators	300, 600, 900
learning_rate	0.03, 0.05 , 0.1
num_leaves	15 , 31, 63
min_child_samples	5 , 10, 20
Best CV R^2 : 0.9905	

Table 21: Final model, 244 universities, evaluated once.

R^2	0.9925
MAE (score points)	1.15
RMSE	1.54
Spearman ρ	0.9918
Kendall τ	0.9298
within ± 1 point	57.0%
within ± 2 points	84.8%
within ± 5 points	99.6%

The search settles on a small tree of 15 leaves with 900 boosting rounds and a low minimum leaf size, which is what one would expect for a target built from many small additive contributions plus a handful of sharp thresholds: shallow trees, many of them, allowed to isolate small groups. These figures should be read subject to Section 5.9, since they measure how well the label is recoverable from the raw attributes rather than how well website quality is predictable in general.

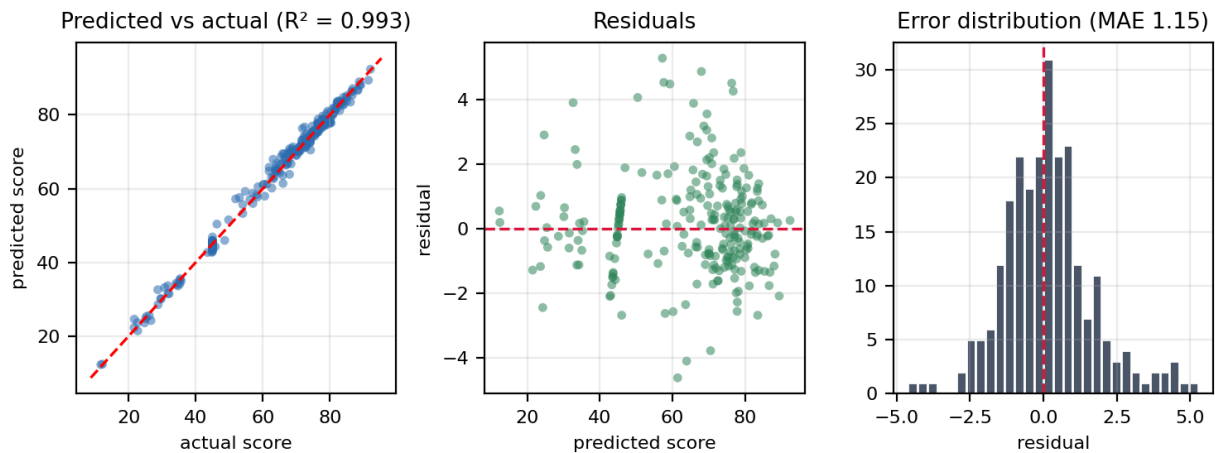


Figure 7: Held-out predictions. Residuals show no systematic pattern with predicted value.

8 Analysis

8.1 Why the Algorithm Families Differ

All twelve algorithms saw identical features and identical training data, yet R^2 ranged from 0.768 to 0.990. Calling that a difference in model capacity explains nothing; the cause is specific and can be demonstrated.

The label contains gates, the hard caps at 45 and 60 points in Equation (3). A linear model computes $\sum_j \beta_j x_j$ and has no way to say “whatever else is true, stop here”. A decision tree does, because a split is exactly that statement. If this is the mechanism, linear models should fail specifically on gated universities and behave normally elsewhere. Splitting the test error confirms it.

Table 22: Mean absolute error on the 35 gated versus 209 ungated test universities.

Model	Family	MAE gated	MAE ungated	ratio
Linear Regression	linear	8.88	1.85	4.80×
Ridge	linear	9.05	1.88	4.81×
Lasso	linear	8.12	1.98	4.11×
SVR (RBF kernel)	kernel	5.73	1.84	3.12×
Neural Net (MLP)	neural	2.46	1.41	1.75×
Decision Tree	tree	0.51	3.86	0.13×
Random Forest	ensemble	0.55	2.36	0.23×
Gradient Boosting	ensemble	1.56	1.54	1.01×
LightGBM	ensemble	0.77	1.40	0.55×

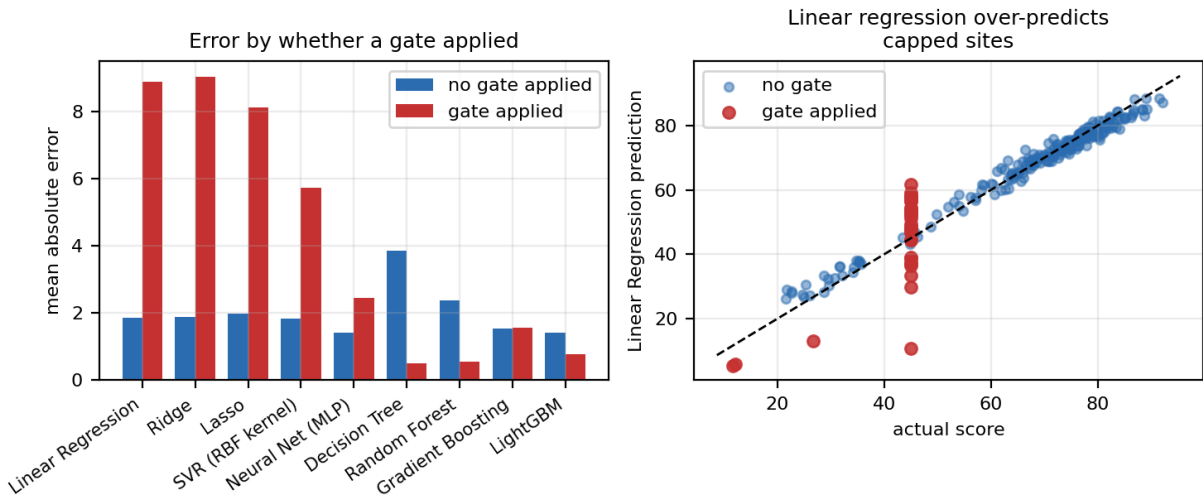


Figure 8: Left: error split by whether a gate applied. Right: linear regression systematically over-predicts the capped universities (red), because it sums the content the site does have and never applies the cap.

Linear regression is wrong by 8.88 points on the gated universities against 1.85 elsewhere, a factor of 4.80. Every tree-based model is as good or better on the gated cases than on the ungated ones, and LightGBM sits at 0.77. The gap between algorithm families is therefore not a vague matter of capacity but the concrete inability of an additive model to represent a threshold, visible on 35 specific universities. The direction of the error supports the mechanism rather than merely correlating with it, since linear regression does not scatter around the gated universities but over-predicts them.

8.2 Feature Importance

We compute importance by gain, meaning how much each split reduced the loss, rather than by split count. The distinction matters: split-count importance flattens continuous attributes, which get split many times, over binary ones, which are split once. Under split-count importance load time looked like the single most important feature; under gain it accounts for well under 1%.

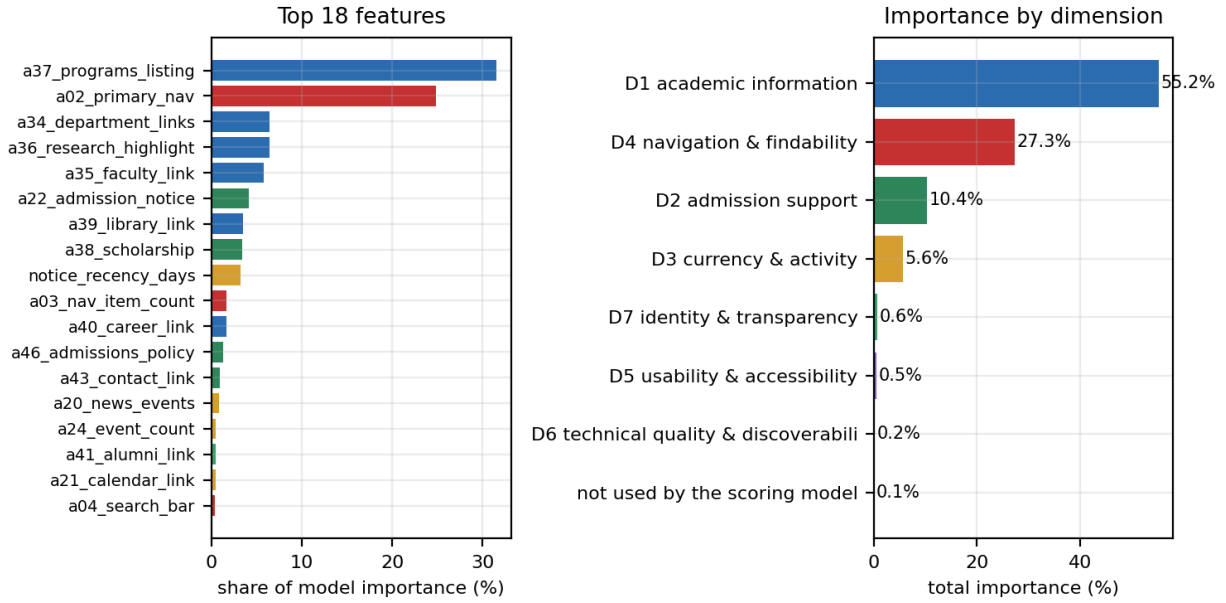


Figure 9: Gain-based feature importance for the tuned model, individually and aggregated by scoring dimension.

Table 23: The fifteen features carrying the most gain in the tuned model.

#	Feature	Gain %	Dimension
1	a37_programs_listing	31.56	D1 academic information
2	a02_primary_nav	24.87	D4 navigation & findability
3	a34_department_links	6.43	D1 academic information
4	a36_research_highlight	6.40	D1 academic information
5	a35_faculty_link	5.74	D1 academic information
6	a22_admission_notice	4.15	D2 admission support
7	a39_library_link	3.47	D1 academic information
8	a38_scholarship	3.37	D2 admission support
9	notice_recency_days	3.23	D3 currency & activity
10	a03_nav_item_count	1.70	D4 navigation & findability
11	a40_career_link	1.64	D1 academic information
12	a46_admissions_policy	1.33	D2 admission support
13	a43_contact_link	0.91	D2 admission support
14	a20_news_events	0.84	D3 currency & activity
15	a24_event_count	0.50	D3 currency & activity

Table 24: Declared dimension weight against realised model importance.

Dimension	Declared weight	Realised importance	Ratio
<i>D1</i> Academic information	28	55.2%	1.97
<i>D2</i> Admission support	22	10.4%	0.47
<i>D3</i> Currency and activity	15	5.6%	0.38
<i>D4</i> Navigation and findability	15	27.3%	1.82
<i>D5</i> Usability and accessibility	10	0.5%	0.05
<i>D6</i> Technical quality and discoverability	7	0.2%	0.03
<i>D7</i> Institutional identity and transparency	3	0.6%	0.21
not used by the label	0	0.1%	—

A declared weight is not the same as realised influence. A dimension can differentiate universities only to the extent that they actually differ on it, and Table 9 showed D_5 to D_7 are close to universal across the population; D_6 drives almost none of the model’s decisions because 98.9% of sites use HTTPS and little is left to explain. As Section 5.9 notes, this is a statement about the dataset rather than a discovery about quality, but it echoes a known result in the website-evaluation literature where a scheme allocating 40% of its score to performance was later found to drive under 3% of its ranking variance.

The model also recovered the gating rule without being told about it. `a02_primary_nav` accounts for 24.9% of all gain, far beyond its arithmetic contribution of 3.75 points inside D_4 , because the model learned that this one binary decides whether a 45-point ceiling applies. Together with `a37_programs_listing` at 31.6%, two of 71 features account for over half of everything the model uses.

8.3 Ablation by Feature Block

Each block of features is deleted from the input and the tuned model refitted on the same training set with the same hyper-parameters. The increase in test MAE is what that block was worth.

Table 25: Block ablation. Blocks are defined twice, once by scoring dimension and once by the page region the attribute came from.

Block removed	Kind	Features	R^2	MAE	Δ MAE
Header & navigation	source group	6	0.675	6.05	+4.90
D4	scoring dimension	6	0.692	6.03	+4.89
Page content	source group	15	0.872	4.69	+3.54
D1	scoring dimension	6	0.936	2.98	+1.83
D2	scoring dimension	7	0.955	2.92	+1.77
Notices & updates	source group	6	0.977	2.00	+0.85
D3	scoring dimension	9	0.981	1.94	+0.79
Events & media	source group	9	0.990	1.37	+0.22
Footer	source group	5	0.991	1.31	+0.16
D5	scoring dimension	6	0.991	1.29	+0.14
Accessibility	source group	4	0.992	1.23	+0.08
SEO & metadata	source group	3	0.992	1.18	+0.04
Visual design	source group	3	0.992	1.17	+0.02
Service & interaction	source group	4	0.993	1.16	+0.01
D7	scoring dimension	10	0.992	1.16	+0.01
D6	scoring dimension	7	0.992	1.16	+0.01
Rankings & recognition	source group	7	0.993	1.16	+0.01
Measurement quality	source group	3	0.993	1.15	+0.00
Technical performance	source group	6	0.992	1.14	-0.01
nothing removed		71	0.993	1.15	—

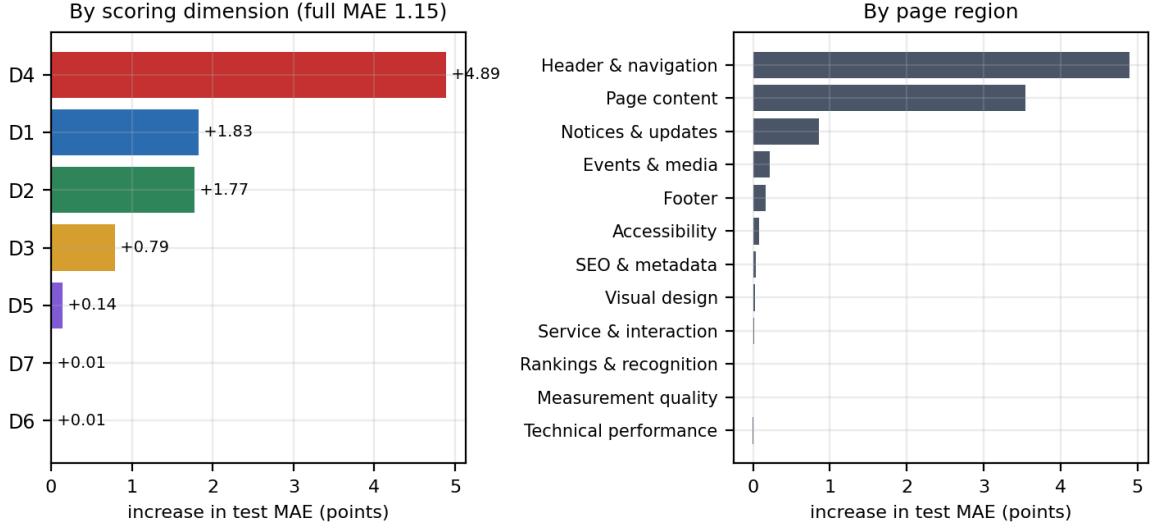


Figure 10: Increase in test MAE when each block of features is removed and the model refitted.

Deleting D_4 costs more than deleting any other block, +4.89 points of MAE against +1.83 for D_1 , even though D_1 carries almost twice the weight in the label. The gate is the reason: without `a02_primary_nav` the model cannot tell which universities are capped at 45, and it loses the sharpest boundary in the data. Section 8.2 reached the same conclusion by a different route.

The blocks the label does not use cost close to nothing. Removing all seven rankings and recognition features, or the three measurement-quality indicators, changes test MAE by a fraction of a point in either direction, so the model correctly identified that these attributes contribute little to the label despite their raw correlation with the score; that is the property Section 5.9 relies on. The two block definitions also act as a consistency check on each other, since header and navigation overlaps heavily with D_4 and costs nearly the same, as it should. Two independent partitions of the feature set agreeing is evidence that the ablation measures something real rather than an artefact of how the blocks were drawn.

8.4 Sensitivity of the Ranking to the Weights

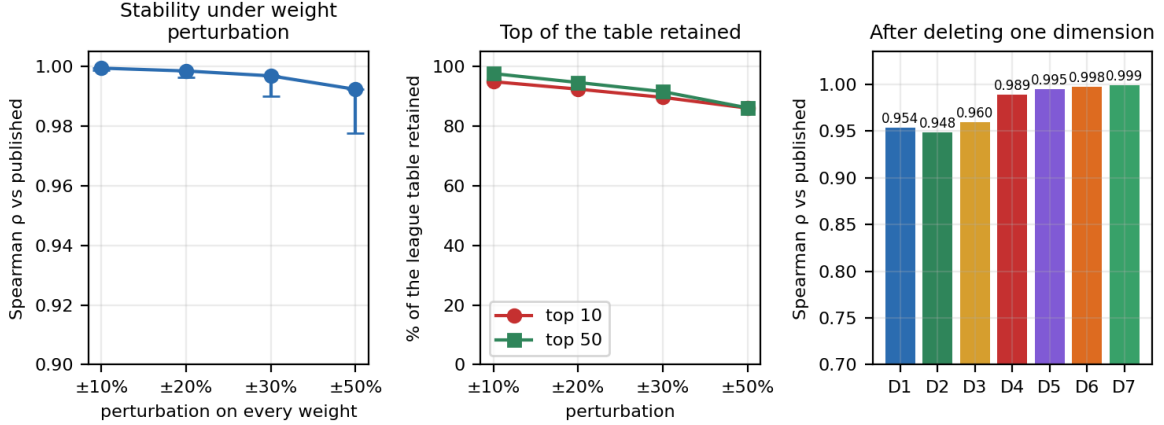
The weights are a documented judgement and a different analyst would choose differently. The honest question is not whether they are correct, since there is no ground truth to be correct against, but how much the league table would move if they were different. Two experiments answer it. In the first, every weight is multiplied by a random factor in $[1 - \lambda, 1 + \lambda]$, renormalised to sum to 100, and the whole ranking recomputed, 300 draws at each of four levels. In the second, each dimension is deleted outright, which is a far more violent change than any plausible disagreement.

Table 26: Ranking stability under random perturbation of all seven weights at once, 300 draws per level. “Top 10 kept” is the fraction of the published top ten still in the top ten of the perturbed ranking.

Perturbation	Spearman ρ	Worst of 300	Top 10 kept	Top 50 kept
± 10				
± 20				
± 30				
± 50				

Table 27: Ranking after deleting one dimension entirely, with the remaining six renormalised to 100.

Dimension deleted	Weight was	Spearman ρ	Top 10 kept	Median rank shift
<i>D1</i> Academic info.	28	0.9537	100%	54
<i>D2</i> Admission support	22	0.9485	40%	45
<i>D3</i> Currency	15	0.9596	80%	48
<i>D4</i> Navigation	15	0.9893	60%	22
<i>D5</i> Usability	10	0.9951	80%	18
<i>D6</i> Technical	7	0.9976	90%	12
<i>D7</i> Identity	3	0.9993	80%	4

**Figure 11:** Left: rank agreement with the published ranking as the weights are perturbed. Centre: membership of the top 10 and top 50 retained. Right: agreement after deleting each dimension outright.

At $\pm 30\%$ on every weight simultaneously, considerably more disagreement than a second analyst would plausibly introduce, the ranking still agrees with the published one at Spearman $\rho = 0.9969$ and keeps 90% of the top ten. Deleting a dimension outright is the only change that moves the table appreciably, and even then it is the high-weight content dimensions that matter; the low-weight ones can be removed with very little effect, which is the correct behaviour for a dimension carrying three points. This is also the one result in the report that does not depend on the label being learnable, since it is a property of the ranking itself.

9 Case Study: Bangladesh

9.1 Protocol and Results

The 22 Bangladeshi universities were removed from the training set by construction (Section 6.1). The model evaluated here was fitted on 981 universities, none of them Bangladeshi, and tuned by cross-validation inside that training set. It is now asked to score all 22. This is the strictest generalisation test in the report: a random 20% test set still lets a model lean on near-neighbours, another campus running the same template, crawled by the same collector, in the same country, and here there are none.

Table 28: All 22 Bangladeshi universities, none seen during training. “World” is the global rank out of 1,225; “Pred. #” is the national position the model would have assigned. KUET is highlighted.

#	University	World	Actual	Predicted	Error	Pred. #	Grade
1	North South Univ.	44	86.82	87.13	+0.31	1	A+
2	RUET	107	83.75	82.56	-1.19	4	A
3	BRAC Univ.	139	82.35	83.75	+1.40	2	A
4	KUET	141	82.18	83.34	+1.16	3	A
5	CUET	173	81.40	80.80	-0.60	6	A
6	BUP	186	81.03	81.84	+0.81	5	A
7	AIUB	201	80.81	79.92	-0.89	7	A
8	PSTU	273	78.57	78.60	+0.03	8	A
9	Khulna Univ.	337	76.98	76.91	-0.07	11	A
10	Daffodil Int. Univ.	344	76.84	77.37	+0.53	9	A
11	Univ. of Barisal	351	76.64	76.95	+0.31	10	A
12	BUTEX	470	74.15	74.91	+0.76	12	B
13	Univ. of Dhaka	502	73.17	73.18	+0.01	13	B
14	Univ. of Chittagong	526	72.64	71.80	-0.84	14	B
15	Comilla Univ.	573	71.28	71.56	+0.28	16	B
16	Jagannath Univ.	576	71.17	71.61	+0.44	15	B
17	Jahangirnagar Univ.	588	70.84	70.07	-0.77	17	B
18	BUET	698	66.80	67.75	+0.95	18	B
19	Univ. of Rajshahi	857	56.15	57.04	+0.89	19	C
20	BAU	932	45.00	45.97	+0.97	20	D
21	IUT	933	45.00	44.08	-0.92	22	D
22	East West Univ.	934	45.00	44.75	-0.25	21	D
<i>mean</i>			71.75	71.90	MAE 0.65		

Table 29: Accuracy on the country holdout, against the rest of the test set.

	Bangladesh (22)	Rest of test set
MAE (score points)	0.65	1.20
R^2	0.9963	—
Spearman ρ	0.9898	—
Largest single error	1.40 points	—
National positions predicted exactly	10 of 22	—

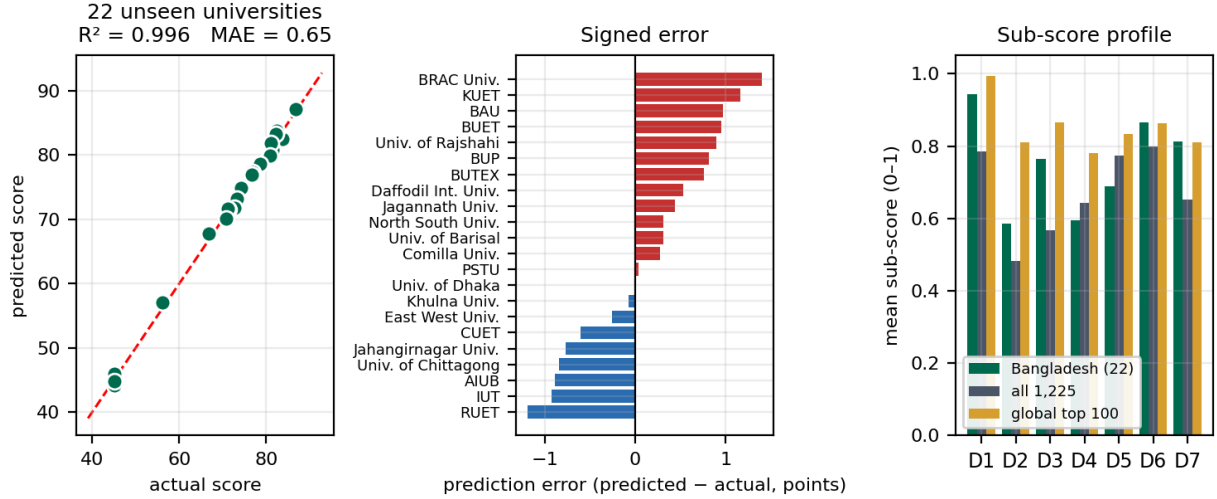


Figure 12: Left: predicted against actual for the 22 unseen universities. Centre: signed error per university, with no systematic bias in either direction. Right: the mean sub-score profile of Bangladeshi websites against the world and against the global top 100.

The MAE on Bangladesh, 0.65, is about half the MAE on the rest of the test set, 1.20, which is the opposite of what a country holdout usually produces. It is not leakage, since the build script asserts there is none, and it is not a paradox. It follows from where these 22 universities sit: 17 of them fall between 71 and 87 points, a dense band the model has seen from a thousand other universities. The model’s errors concentrate at the extremes of the score range, and only three Bangladeshi sites are down there, the three that hit the navigation gate, all of which the model places correctly near 45. We read this as showing that the scoring structure learned elsewhere transfers to a country the model never trained on, and that Bangladeshi university websites are structurally ordinary rather than a distinct population needing their own model. It does not show that the model would transfer as well to a country structurally unlike anything in the training set.

For a league table the ordering matters more than the score, and the model reproduces the national ordering at Spearman $\rho = 0.9898$. The remaining disagreements are local rather than structural, occurring inside tightly packed groups where several universities sit within about a point of each other, which is the same order as the model’s own error, and no university crosses more than one grade boundary. The top of the national table, the middle group and the gated tail are all placed correctly; the exact ordering within a cluster of near-identical sites is not something a model with roughly one point of error can be expected to resolve.

9.2 National Dimension Profile

Multiplying each dimension gap in Figure 12 by that dimension’s weight converts it into points and locates the difference between an average Bangladeshi site and a global top-100 site.

Table 30: Mean sub-score of the 22 Bangladeshi universities against the full dataset and against the global top 100. The last two columns convert the gap into points of final score.

Dimension	BD	World	Top 100	vs world	vs top 100
<i>D1</i> Academic info. ($w = 28$)	0.943	0.785	0.993	+4.42	-1.40
<i>D2</i> Admission support ($w = 22$)	0.585	0.483	0.811	+2.24	-4.97
<i>D3</i> Currency ($w = 15$)	0.765	0.568	0.865	+2.96	-1.50
<i>D4</i> Navigation ($w = 15$)	0.594	0.643	0.780	-0.74	-2.79
<i>D5</i> Usability ($w = 10$)	0.688	0.773	0.833	-0.85	-1.45
<i>D6</i> Technical ($w = 7$)	0.866	0.799	0.862	+0.47	+0.03
<i>D7</i> Identity ($w = 3$)	0.813	0.652	0.811	+0.48	+0.01
Total (points)				+8.98	-12.07

Bangladeshi university websites average 71.8 points against a world mean of 63.9, comfortably above average, and 6 of the 22 reach the global top 200. Their strength is content: on D_1 they score 0.943 against a world mean of 0.785, close to the global top 100, and they publish their programmes, departments and faculty well.

The weakness is not the content but access to it. D_4 , navigation and findability, is the only dimension where Bangladeshi sites fall below the world average, 0.594 against 0.643, and it is worth -2.79 points against the top 100. D_2 , admission support, is the largest single gap at -4.97 points. Together those two account for 7.76 of the 12.1-point gap to the global top 100. The practical conclusion is encouraging: these universities have already done the expensive work of putting their academic content online, and what separates them from the global leaders is a menu, a search box, a breadcrumb trail and a clearly linked admissions policy. Three institutions, Bangladesh Agricultural University, East West University and the Islamic University of Technology, have no extractable primary navigation at all and are capped at 45, forfeiting 17.6, 25.9, 29.2 points respectively.

9.3 Prestige and Website Quality

Table 28 contains a result worth stating plainly, because it is the clearest justification for excluding the external ranking values in Section 3 and the ranking badges in Section 5.7. The Bangladesh University of Engineering and Technology, by any conventional measure the country’s most prestigious institution, ranks 18th of 22 on website quality with a score of 66.80 and a grade of B, while a private university founded in 1992 leads the national table. Institutional prestige and landing-page quality are close to unrelated in this dataset, which is why a website-quality ranking is worth computing separately rather than assuming existing rankings already answer the question.

10 Rankings

Table 31: Global and national leaders.

Global top 10			Bangladesh top 8 (of 22)			
#	University	Score	#	University	Global	Score
1	Eotvos Lorand Univ. (HU)	94.9	1	North South Univ.	44	86.8
2	Univ. of New England (AU)	94.4	2	RUET	107	83.8
3	Univ. of Strasbourg (FR)	93.7	3	BRAC Univ.	139	82.3
4	Univ. of Porto (PT)	93.4	4	KUET	141	82.2
5	Nanyang Tech. Univ. (SG)	92.9	5	CUET	173	81.4
6	Gauhati Univ. (IN)	92.7	6	BUP	186	81.0
7	RWTH Aachen Univ. (DE)	92.5	7	AIUB	201	80.8
8	Univ. of Leicester (UK)	92.1	8	PSTU	273	78.6
9	Southern Cross Univ. (AU)	92.0				
10	Maynooth Univ.	91.9				

Country rankings are produced for the 18 countries with at least 20 universities in the dataset, covering 779 universities. For the 169 rows whose `country` field holds a regional label rather than a country we produce no country ranking, because it would not be one. The full Bangladesh table, every position of which is an out-of-sample prediction, is Table 28.

11 Risks and Limitations

Table 32: The principal risks to the validity of this study, and how each is handled.

Risk	Impact	Disposition
Label leakage: the label is an arithmetic function of the features	Reported accuracy is circular if read as quality prediction	Not eliminated. The research question is narrowed accordingly and the limitation stated in full in Section 5.9. Sub-scores, curve outputs and the gate flag are withheld from the model, the 20 unscored features are supplied as distractors, and the ranking is validated independently of the model in Section 8.4
Prestige leakage via displayed QS / Webometrics values	The model ranks universities rather than web-sites	Ranking value columns dropped in cleaning, being 95–99% null and leaky; the badge presence flags are retained as page features but excluded from the label, and ablation shows the whole block is worth almost nothing
Inconsistent extraction across six collectors	Systematic between-region offsets mistaken for quality differences	Collector and region are perfectly confounded and this cannot be undone; load time is used only in region-standardised form, and the confound is reported rather than adjusted for
JavaScript-heavy sites under-extracted	Content attributes systematically undercounted	About 2% of rows affected; they fall to the bottom of the ranking and are indistinguishable from genuinely un navigable sites

1. **The label is a function of the features**, as set out in Section 5.9. It bounds every accuracy figure in Sections 7 and 7.2 to a claim about recoverability rather than about quality prediction.
2. **The dataset cannot see visual design.** Every attribute is a presence flag, a count or a measurement, so whether a page is cluttered, dated-looking or unpleasant to use is not among the 71 columns and cannot be in the label. This is also why seven media and visual attributes

are excluded from it. Closing this gap needs screenshots and human raters rather than more modelling.

3. **About 2% of rows are extraction failures.** A small number of universities returned almost nothing to the crawler, which is characteristic of JavaScript-rendered pages the extractor could not read rather than of genuinely empty websites. Since the navigation gate is triggered by exactly this failure mode, an extraction failure and a genuinely unnavigable site look the same to the label, so the bottom of the ranking should not be quoted without this caveat.
4. **The score reflects the landing page,** not the whole site and not the institution. An excellent admissions portal that the front page does not link to is invisible to this method.
5. **The weights are a documented judgement, not a measured truth.** A stated rule fixed them before scoring and they are published in full, and Section 8.4 quantifies the consequence: at $\pm 30\%$ on every weight the ranking holds at $\rho = 0.9969$.
6. **Excluding 20 attributes is itself a judgement.** Each exclusion has a stated reason (Table 14), but reasons can be disagreed with. The most arguable is `a28_contests`, excluded as an engagement signal rather than an information need despite correlating with the score at 0.29.
7. **The Bangladesh result is one country,** not a general claim about transfer. The 22 Bangladeshi sites resemble their South Asian neighbours, of which the training set contains many.
8. **Region and data collector are perfectly confounded,** so regional differences in a measurement such as load time cannot be separated from differences in how it was measured.
9. **The crawl is a single snapshot.** Websites change continuously, the freshness attributes in particular would differ on another day, and D_3 is the least stable dimension in the label over time.

12 Conclusion

We collected and audited a global corpus of 1,225 landing pages described by 71 attributes, classified every attribute by page region, measurement type and scoring role, and built a seven-dimension label grounded in what a prospective student needs, derived by task decomposition and stated as equations with every inner weight, response curve and gating rule published, along with a reason for each of the 20 attributes the label does not use. Twelve regression algorithms were then trained to predict that label from the raw attributes. The tuned LightGBM model reaches $R^2 = 0.9925$ with a mean absolute error of 1.15 points on a test set held out from every fitting and tuning decision, ordering the universities at Spearman $\rho = 0.9918$.

The comparative study is the more instructive half. Performance across the twelve algorithms was not a smooth gradient of capability but a split along one structural line: models that can represent a threshold and models that cannot. Isolating the 35 gated universities made the mechanism visible, with linear regression wrong by 8.88 points on exactly those cases against 1.85 elsewhere while LightGBM sits at 0.77, and gain-based importance confirmed that the winning model had independently identified the gating attribute and given it 24.9% of all loss reduction. The ablation study reached the same conclusion from the opposite direction: deleting the six navigation features costs more test accuracy than deleting the six academic ones, despite academic information carrying nearly twice the declared weight.

The country holdout is the strongest evidence that the framework generalises. All 22 Bangladeshi universities were excluded from training by construction, and on them the model achieves MAE = 0.65 and reproduces the national ordering at $\rho = 0.9898$ with a worst-case error of 1.40 points. The case study also yields an actionable diagnosis: Bangladeshi university websites sit above the world average and are strong on academic content, and what separates them from the global top 100 is navigation and admission support, worth 7.76 of the 12.1-point gap.

Three methodological points generalise beyond this dataset. Missing values should be filled according to what absence means rather than by a default statistic, since median-filling one attribute here would have described 468 abandoned notice boards as updated yesterday. A declared weight is not a realised influence: the three lowest-weighted dimensions account for a negligible share of what the trained model uses, simply because almost every university scores alike on them. And a subjective design choice is not a reason to leave it unmeasured, since the weights are a judgement but the sensitivity of the ranking to that judgement is an experiment we could run, and did.

The limitation of Section 5.9 should be carried out of this report along with its results. The label is an arithmetic function of the features, so the accuracy figures measure recoverability rather than quality prediction. What the study establishes is narrower than the headline number suggests and, because the mechanism behind the spread between algorithm families is identified rather than merely reported, more durable.

12.1 Future Work

1. **Replace the constructed label with an independent one.** Human ratings on a stratified subset, or blind pairwise judgements fitted with a Bradley–Terry model, would give a target that is not a function of the features and would turn the accuracy figures from recoverability into genuine prediction. This is the highest-value extension and addresses Section 5.9 directly.
2. **Add visual features.** Screenshot-derived attributes and a small panel of human raters would address limitation 2 and would let the seven media and visual attributes currently excluded from the label be scored on quality rather than presence.
3. **Replace the extractor with a headless browser** to eliminate the JavaScript rendering failures and with them the confusion between an unnavigable site and an unreadable one.
4. **Generalise the country holdout.** Extending the protocol of Section 9 to every country with at least 20 universities would turn a single case study into a leave-one-country-out generalisation study.

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